

Lesson Six

Execute & Close of software project

Effective Project Management: Traditional, Agile, Extreme

Ch09: How to Execute a TPM Project

Ch10: How to Close a TPM Project

***Presented by
Bowen DU***

Lesson Six: Execute & Close of software project

Outline

- Establishing your progress reporting system.
- Applying graphical reporting tools.
- Building and maintaining the Issues Log.
- Managing project status meetings.
- Writing & maintaining client acceptance procedures
- Closing a project.
- Getting client acceptance.
- Documenting the project.
- Conducting the post-implementation audit.
- Writing the final report.

Lesson Six: Execute & Close of software project

Characteristics of Effective Progress Reporting

- Timely, complete, accurate, and intuitive.
- Isn't burdensome and counterproductive.
- Readily acceptable to senior management.
- An effective early warning system.
- Easily understood by those who need to know.

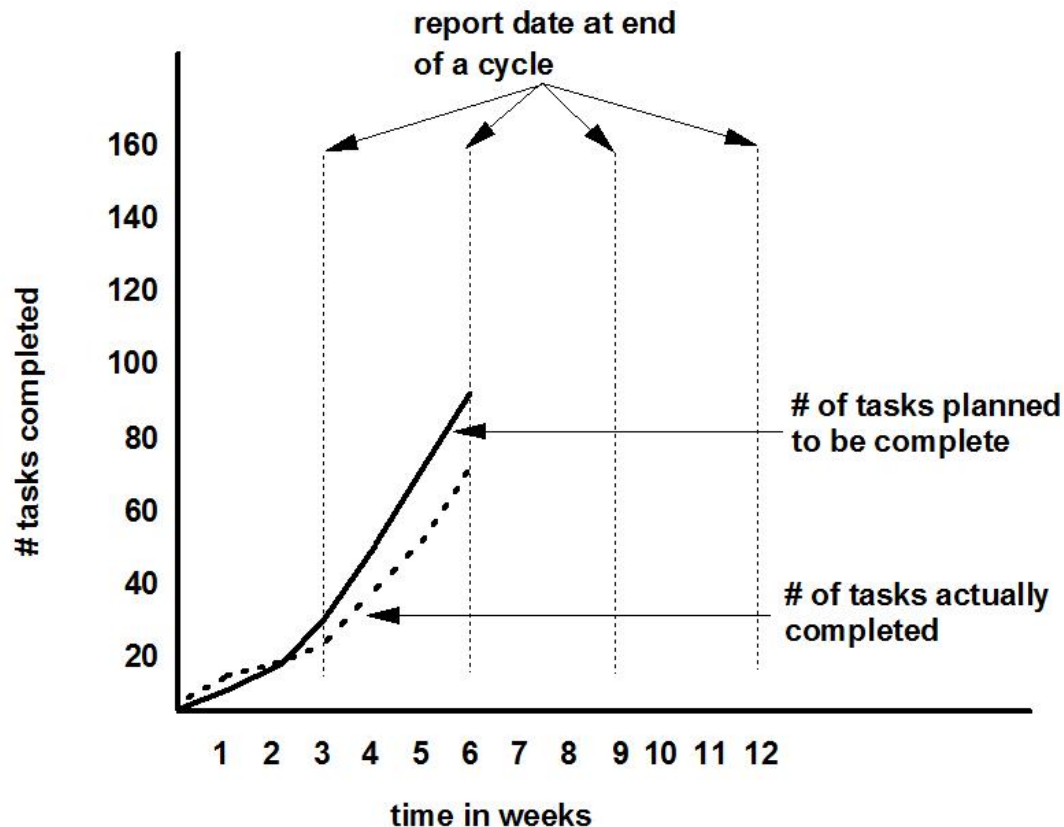
Lesson Six: Execute & Close of software project

Five Types of Project Status Reports

- Current period reports.
- Cumulative reports.
- Exception reports.
- Stoplight reports.
- Variance reports.

Lesson Six: Execute & Close of software project

Gantt Chart Project Status Report



Lesson Six: Execute & Close of software project

Why Measure Duration and Cost Variances

- Catch deviations from the curve early.
- Dampen oscillation(抑制振荡).
- Allow early corrective action.
- Determine weekly schedule variance.
- Determine weekly effort (e.g. person hours/day) variances.

Lesson Six: Execute & Close of software project

How & What Information to Update

- Determine a set period of time and day of week.
- Report actual work accomplished during this period.
- Record historical and re-estimate remaining.
- Report start and finish dates.
- Record days of duration accomplished and remaining.
- Report resource effort spent and remaining.
- Report percent complete.

Lesson Six: Execute & Close of software project

Frequency of gathering & reporting project progress

- According to your software process.
- A logical frequency for reporting is once a week.
- Let people extrapolate(推测) to the end of the workday.

Lesson Six: Execute & Close of software project

How to Keep a Project on Schedule

- Hold daily team meetings.
- Complete tasks ASAP.
- Report problems ASAP.
- Don't fall victim to the "creeps".
- Don't guess – ask questions.
- Good enough is good enough.
- Meet but do not exceed requirements.
- Be open and honest with your team mates.

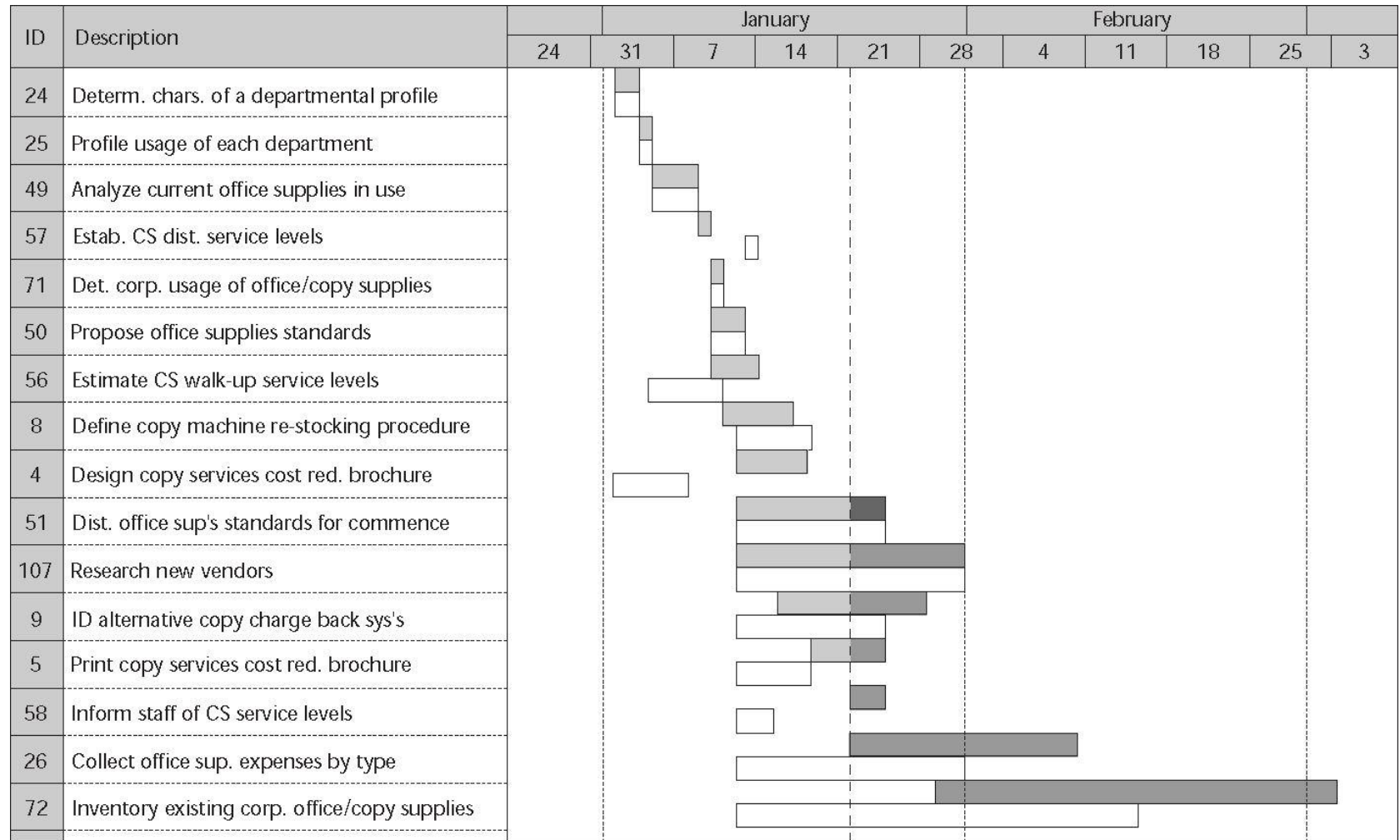
Lesson Six: Execute & Close of software project

Variances

- **Positive Variances** – deviations from the plan indicating that an ahead-of-schedule or a cost less than budgeted has occurred.
- **Negative Variances** – deviations from the plan indicating that a behind-schedule or cost greater than the budgeted has occurred.

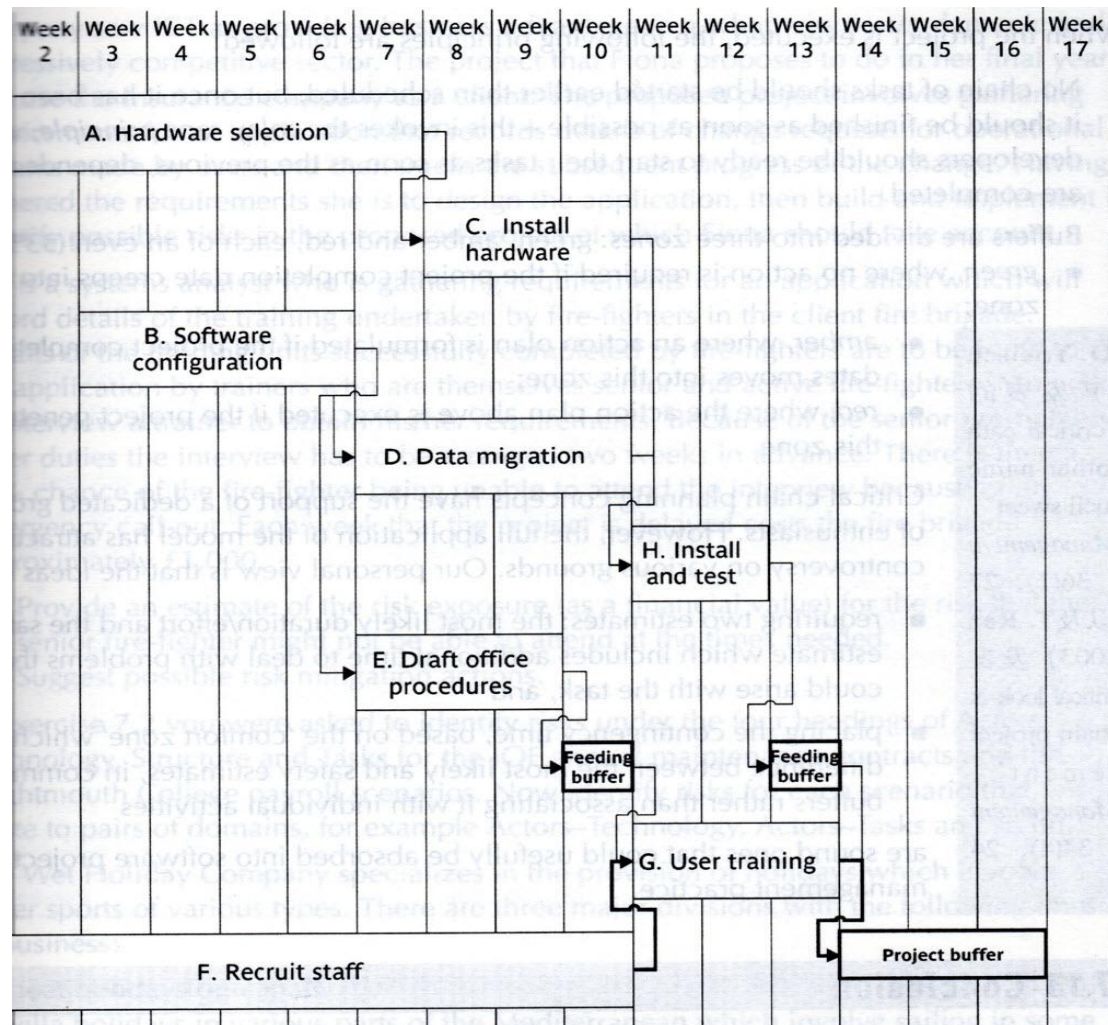
Lesson Six: Execute & Close of software project

Gantt Chart Project Status Report

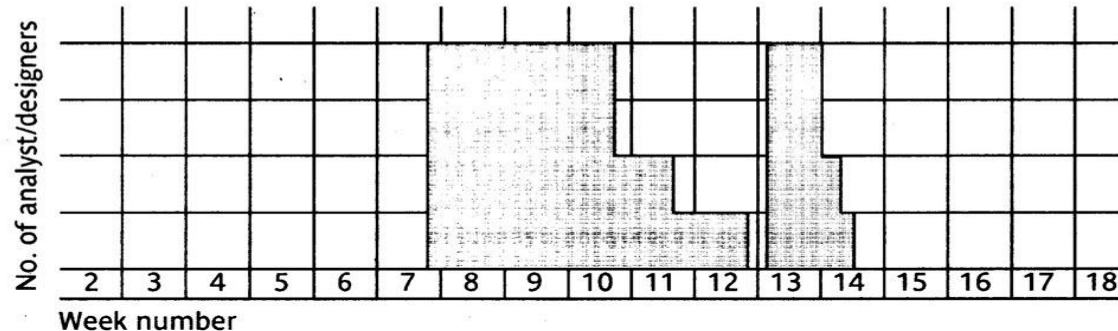


Lesson Six: Execute & Close of software project

Example – Gantt Chart

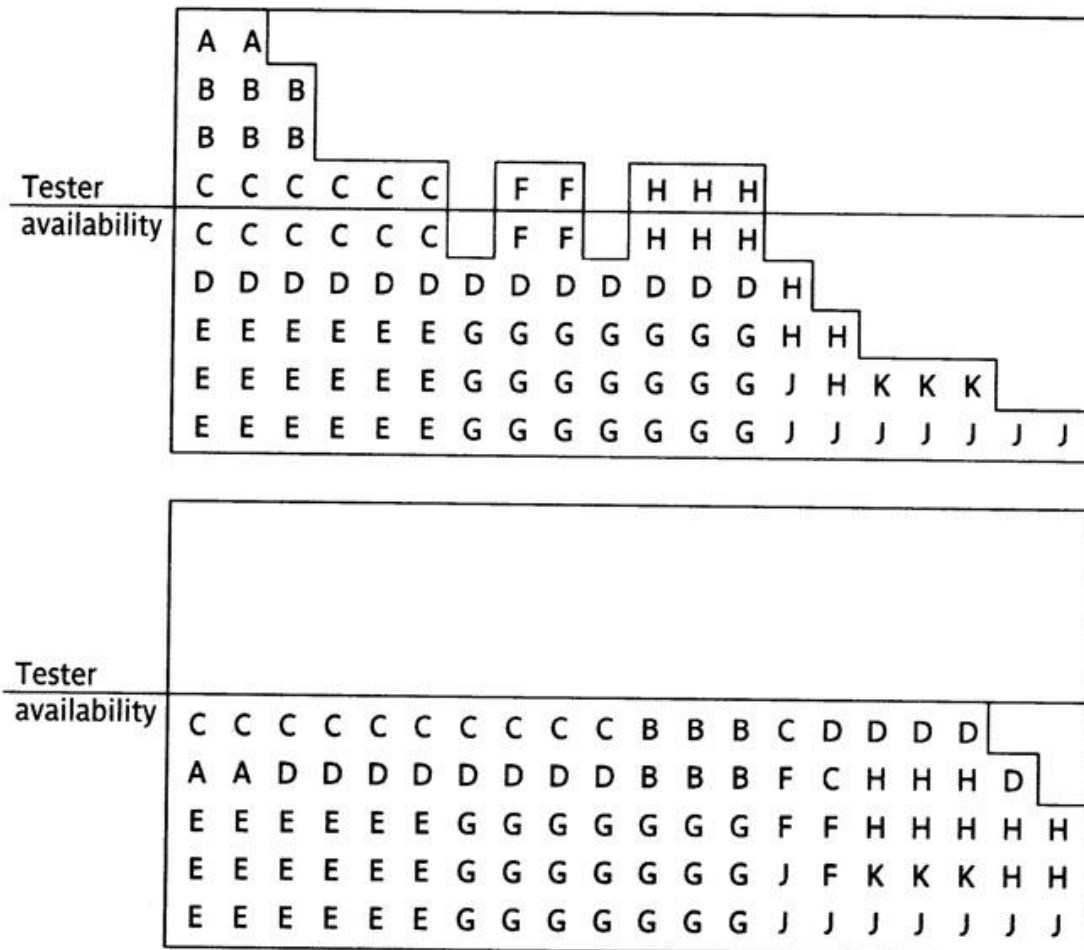


Example – Gantt Chart



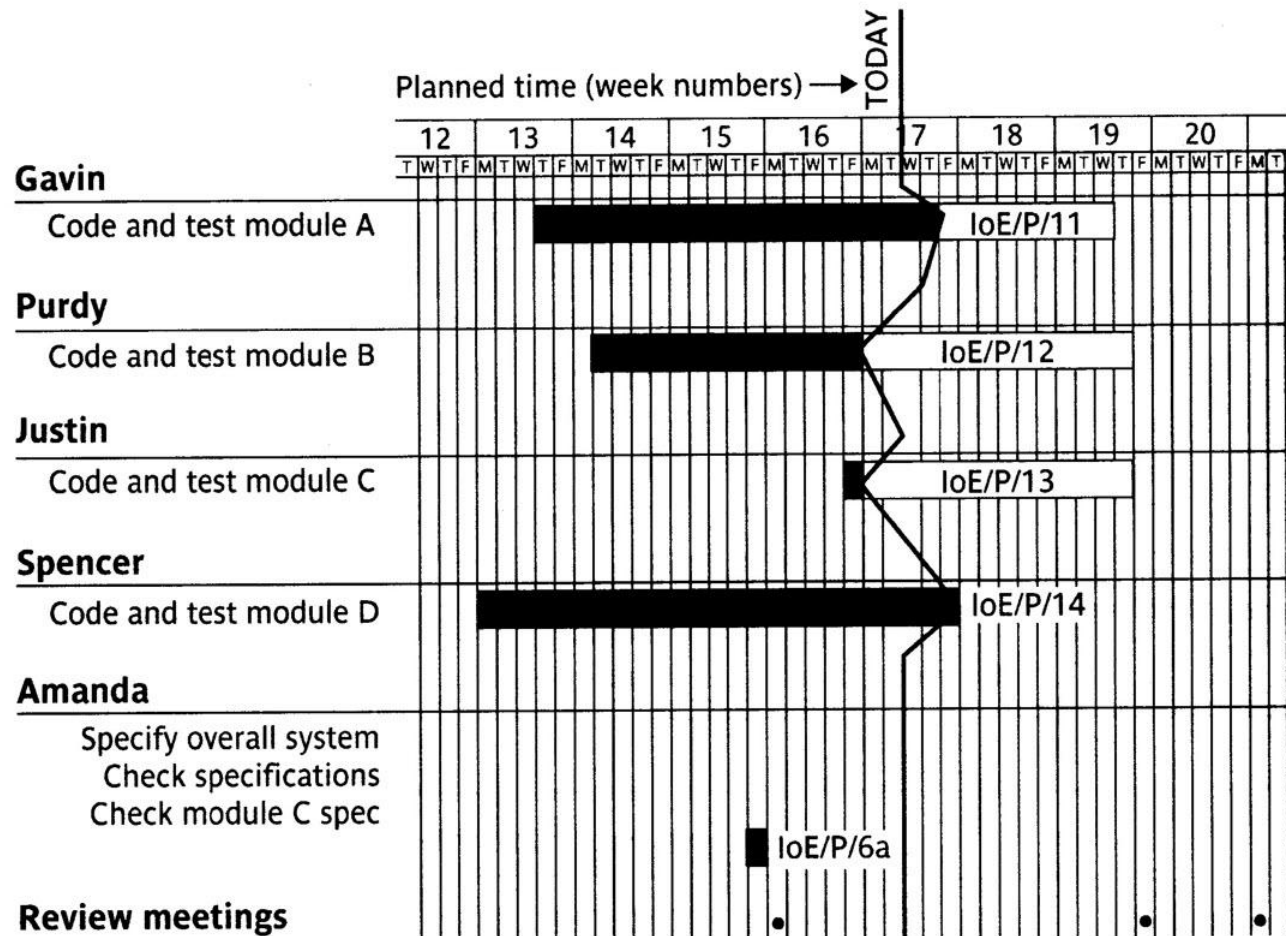
Lesson Six: Execute & Close of software project

Example – Gantt Chart



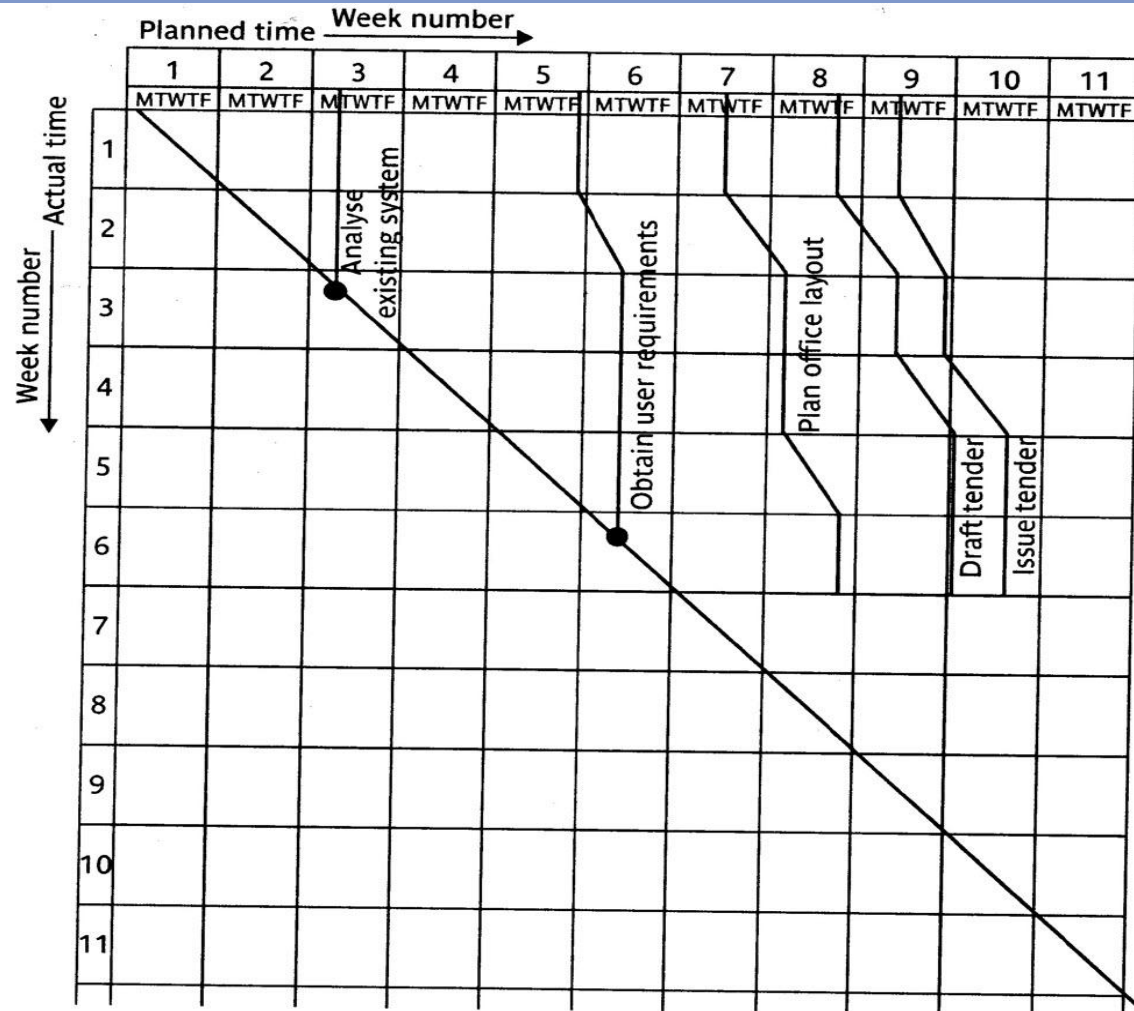
Lesson Six: Execute & Close of software project

Example – Gantt Chart



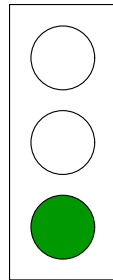
Lesson Six: Execute & Close of software project

Example – Gantt Chart

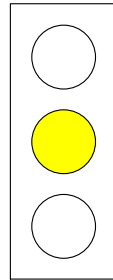


Lesson Six: Execute & Close of software project

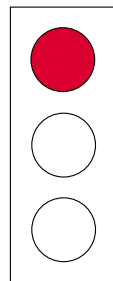
Exception Report – Stoplight Reports



The project is progressing according to plan.



**The project has a problem.
A Get Well plan is in place.
The situation will correct.**



**The project is failing.
Intervention is required.**

Lesson Six: Execute & Close of software project

Exception Report – Stoplight Reports

Activity Assessment Sheet

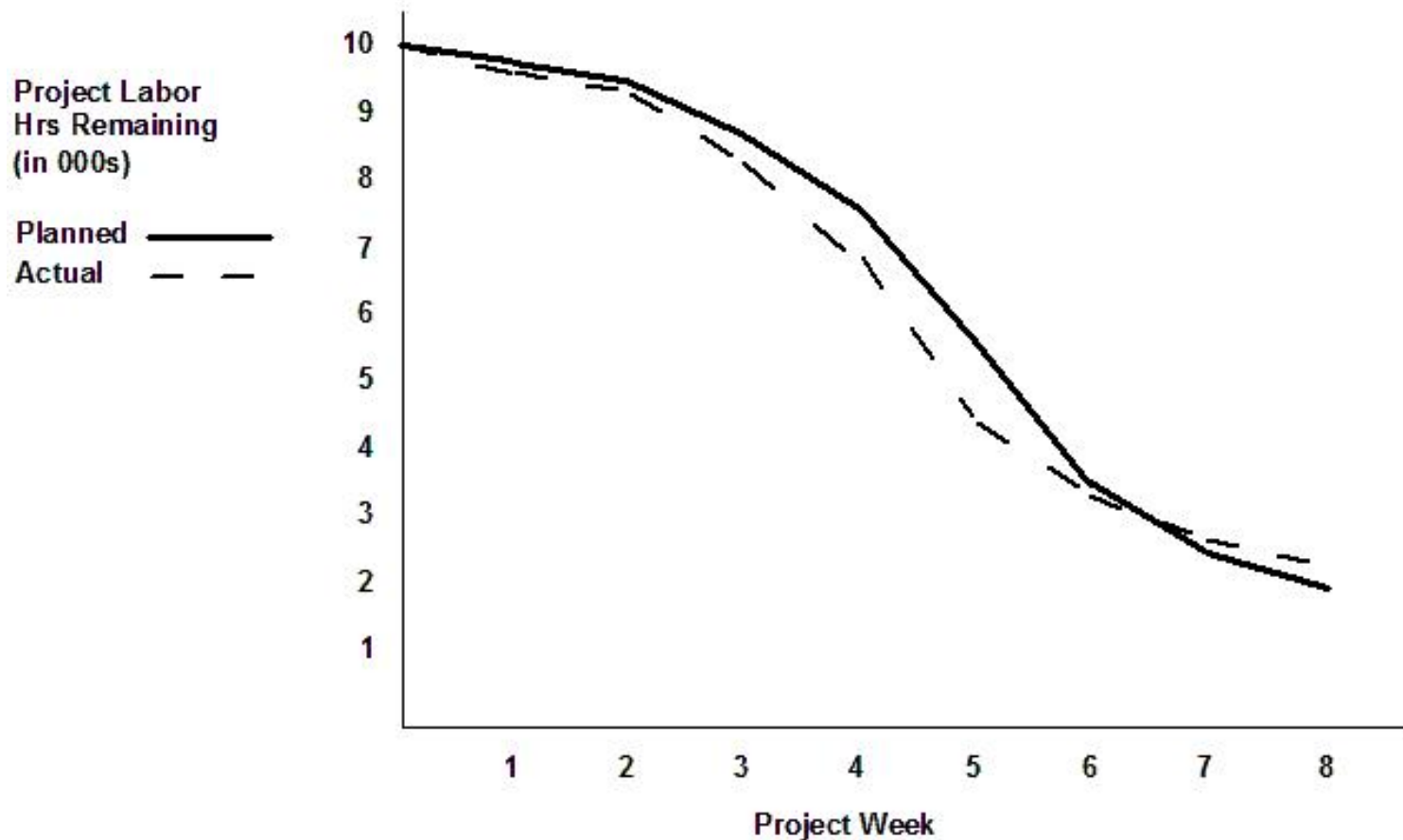
Staff Justin

Ref: IoE/P/13 Activity: Code and test module C

Week number	13	14	15	16	17	18	
Activity summary	G	A	A	R			
Component							Comments
Screen handling procedures	G	A	A	G			
File update procedures	G	G	R	A			
Housekeeping procedures	G	G	G	A			
Compilation	G	G	G	R			
Test data runs	G	G	G	A			
Program documentation	G	G	A	R			

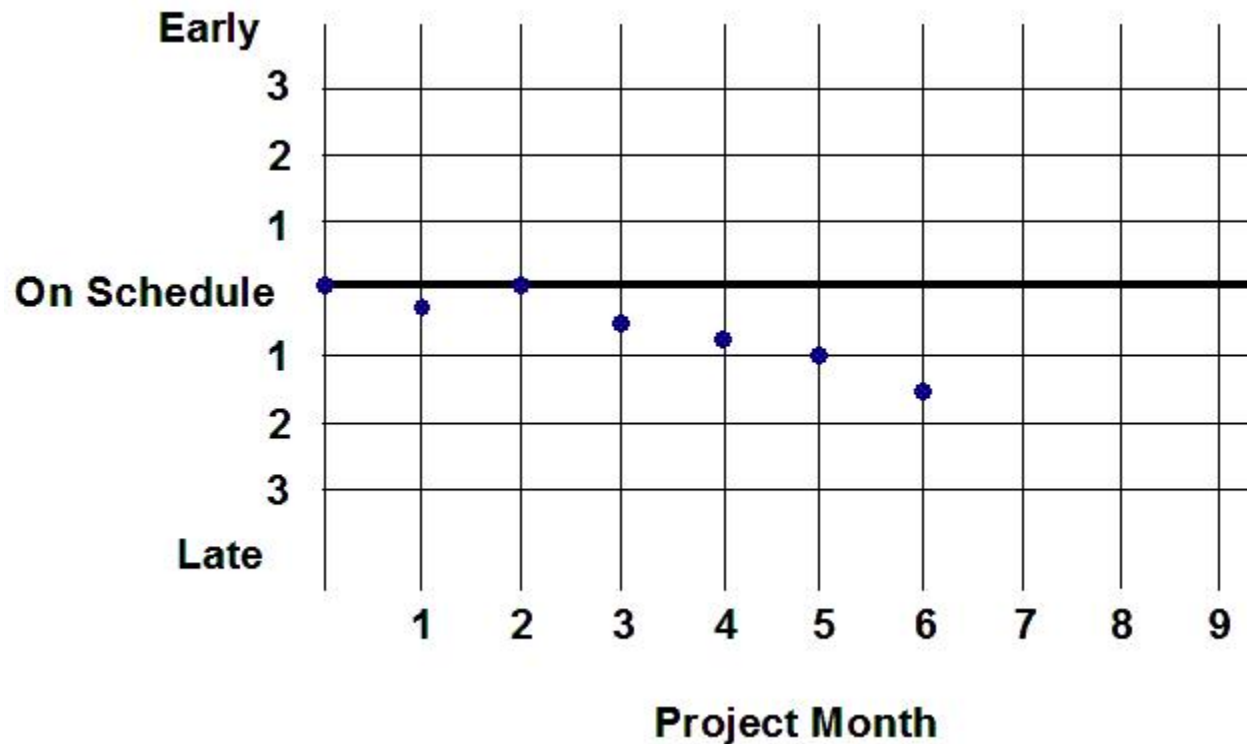
Lesson Six: Execute & Close of software project

Burn Charts



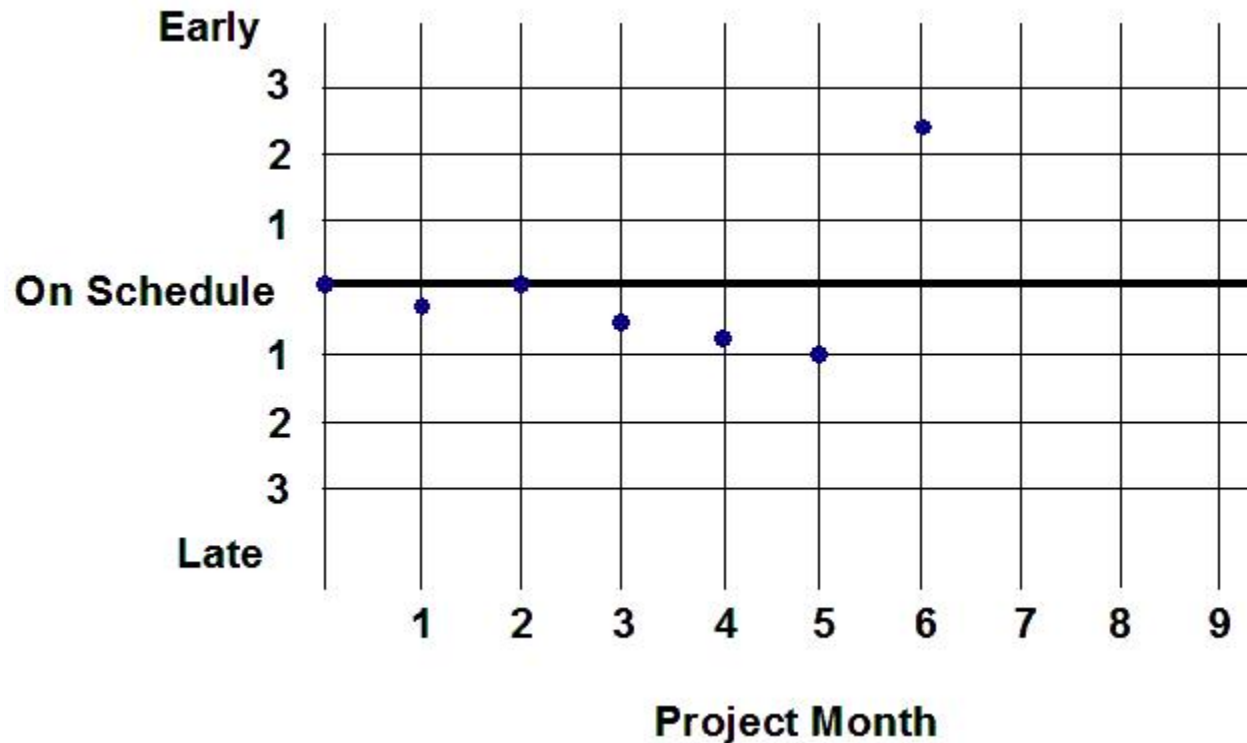
Lesson Six: Execute & Close of software project

Cumulative Reports - Milestone Trend Charts



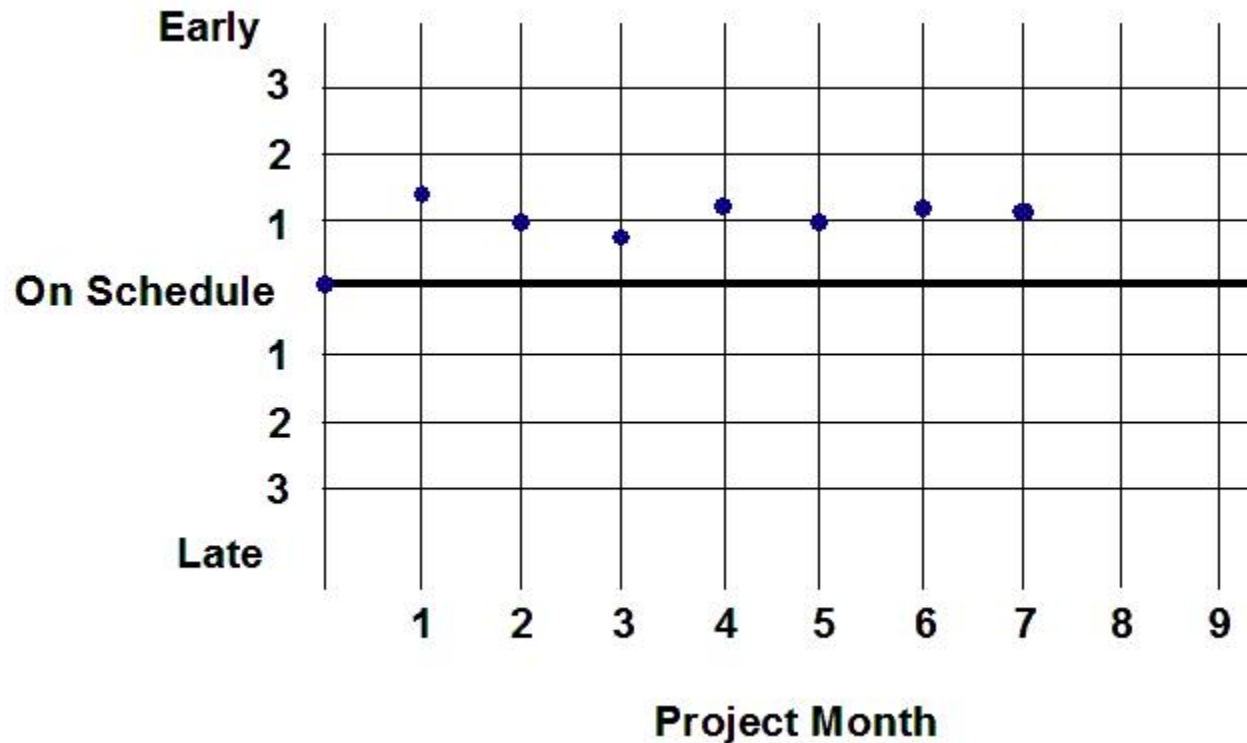
Lesson Six: Execute & Close of software project

Cumulative Reports - Milestone Trend Charts



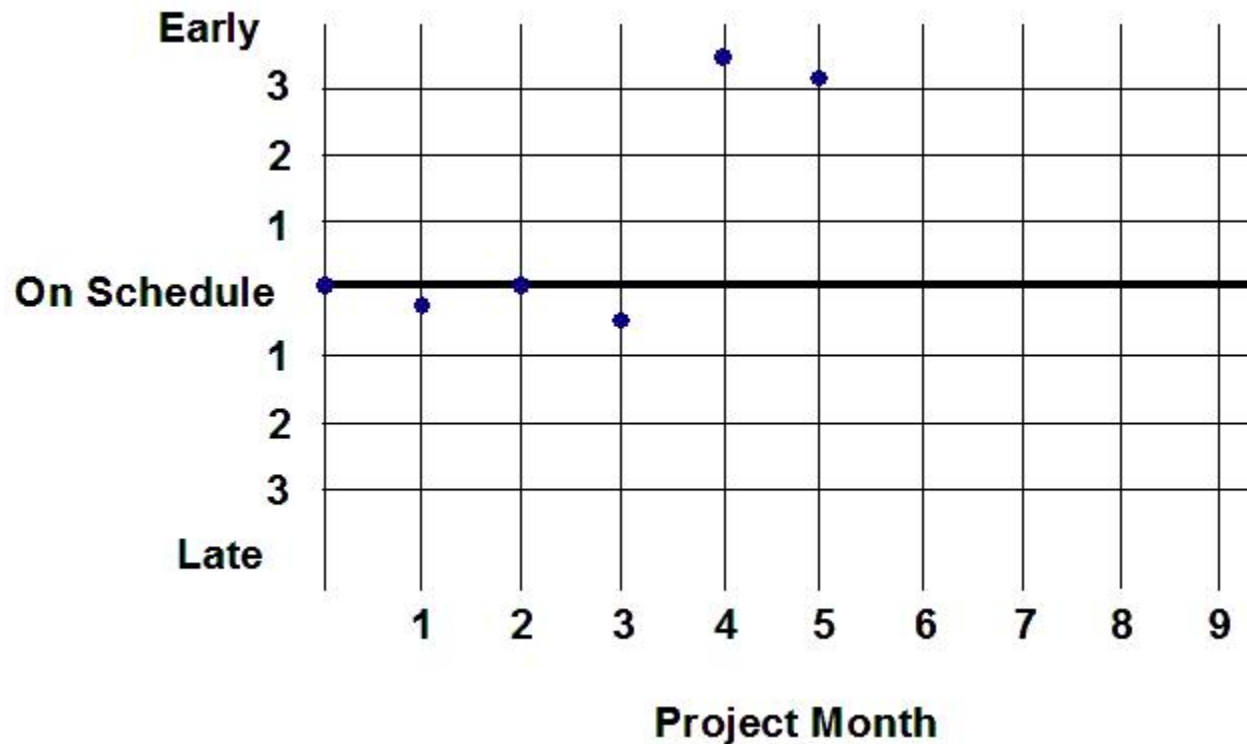
Lesson Six: Execute & Close of software project

Cumulative Reports - Milestone Trend Charts



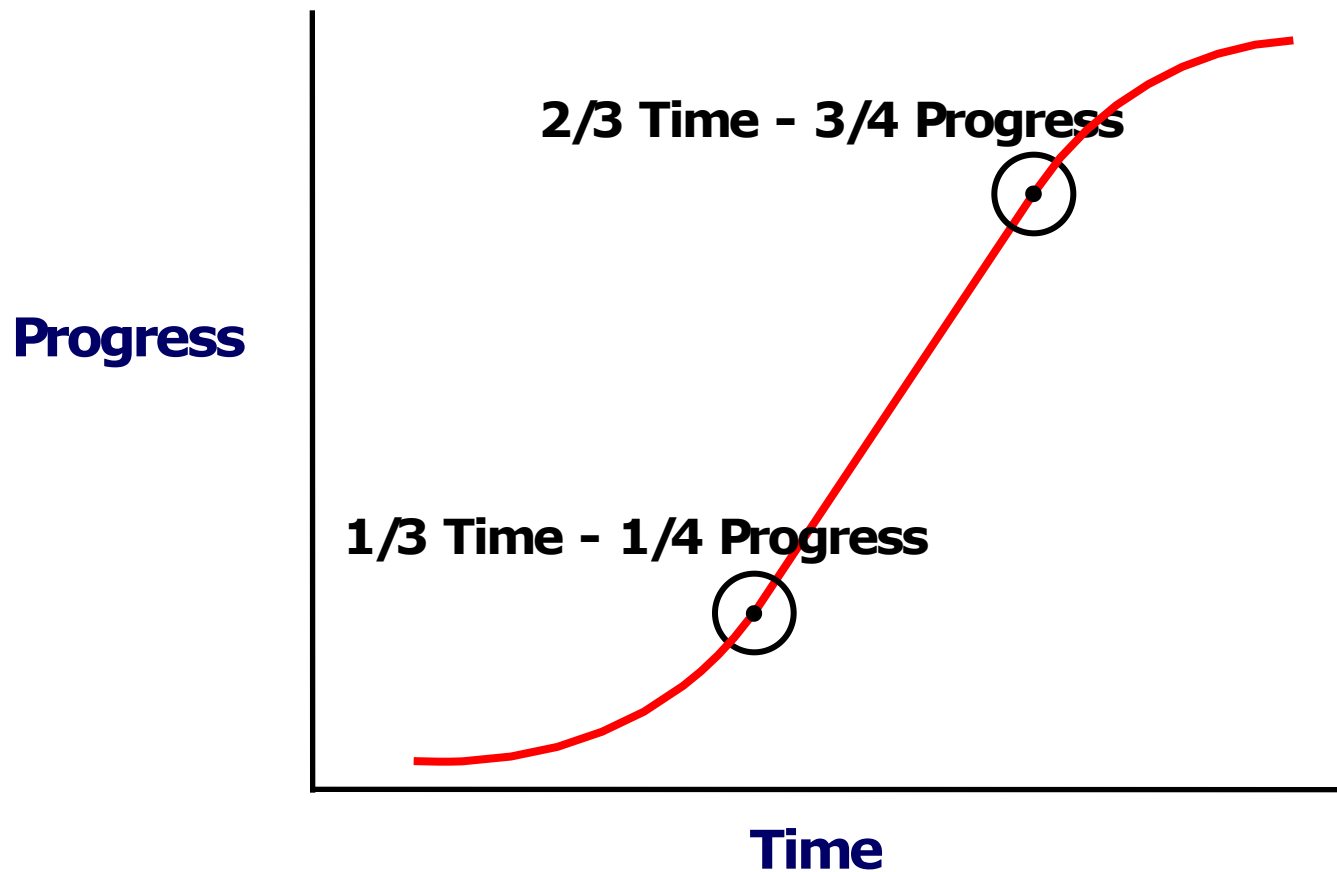
Lesson Six: Execute & Close of software project

Cumulative Reports - Milestone Trend Charts



Lesson Six: Execute & Close of software project

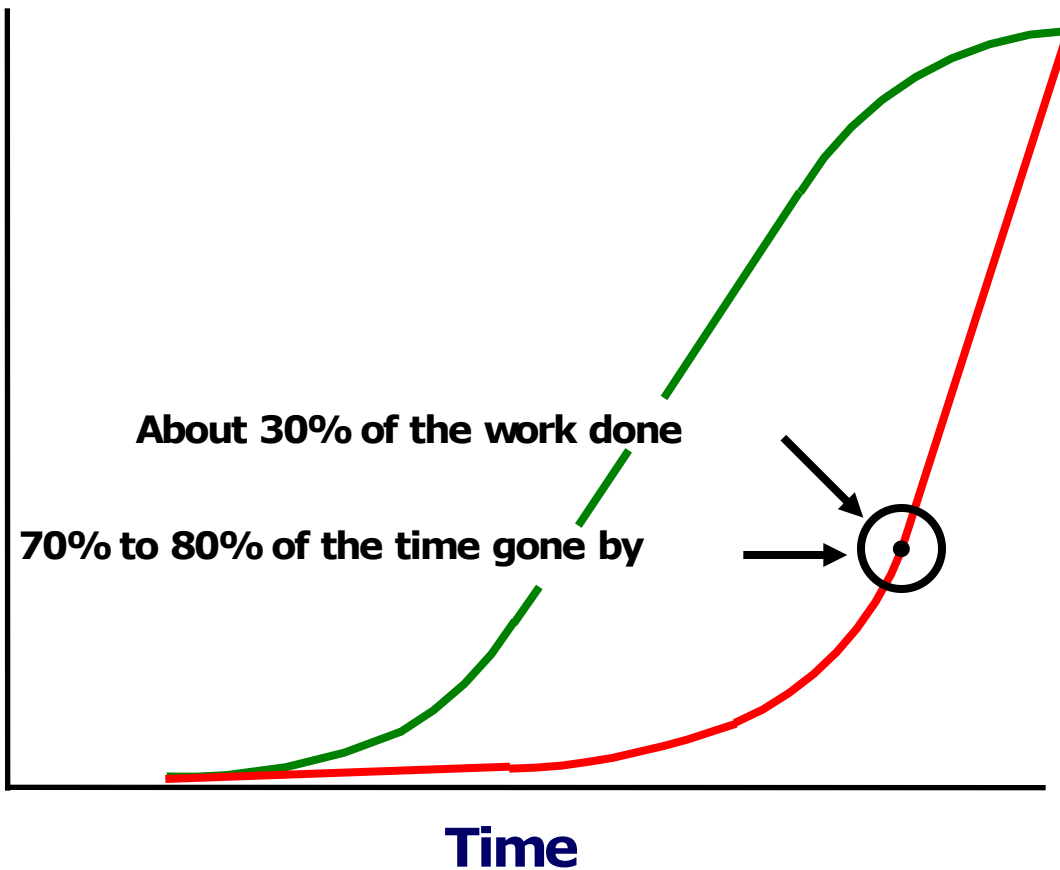
Earned Value – The Standard S-Curve



Lesson Six: Execute & Close of software project

Earned Value – The Curve to Avoid

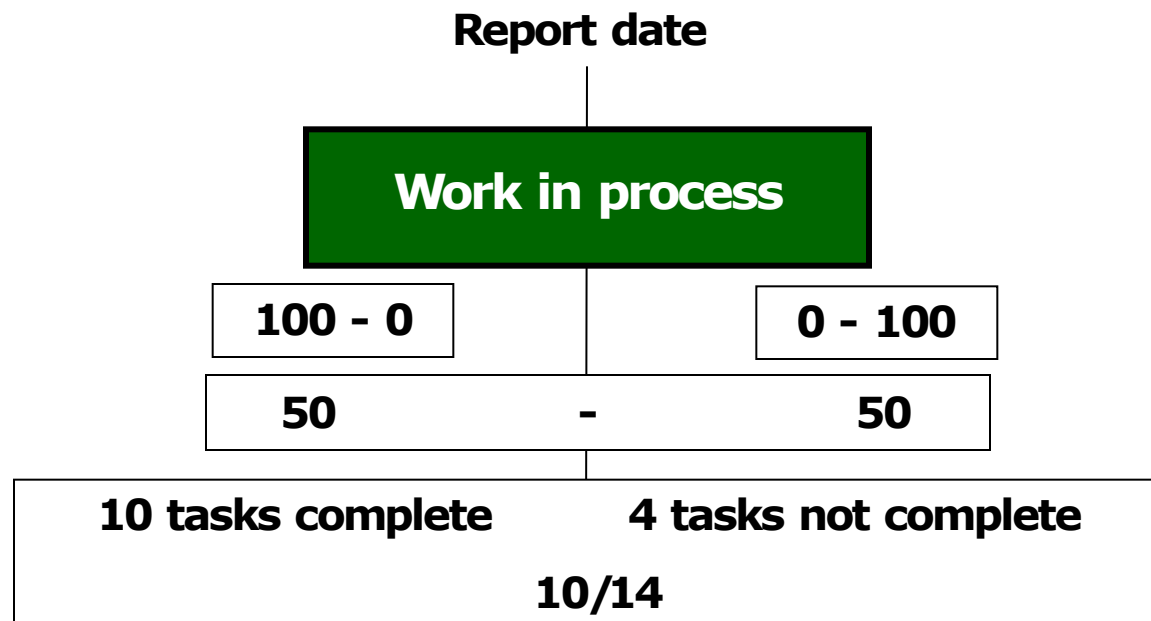
Progress



Lesson Six: Execute & Close of software project

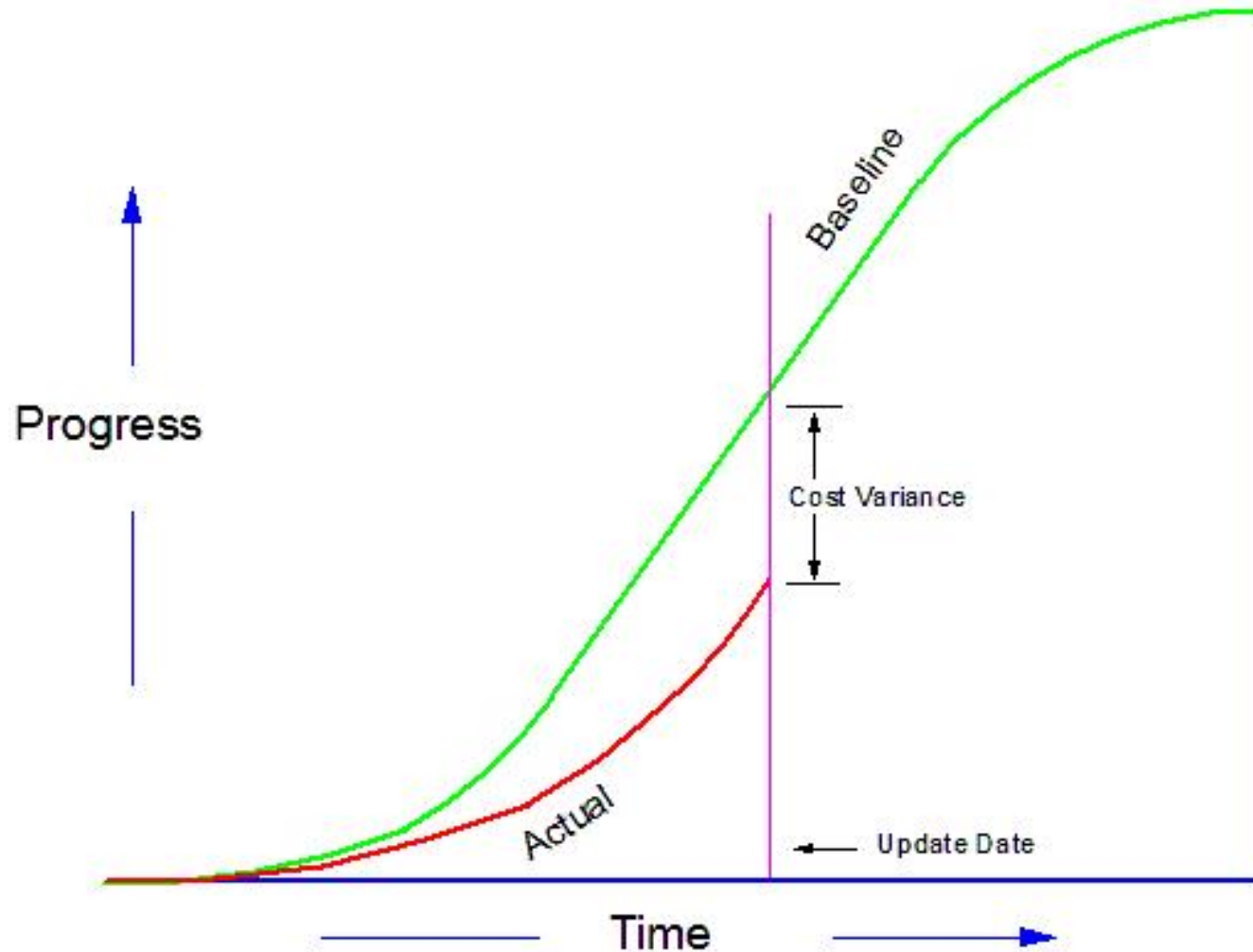
How to Measure Percent of Value Earned

- $100 - 0$
- $0 - 100$
- $50 - 50$
- Proportion of tasks completed



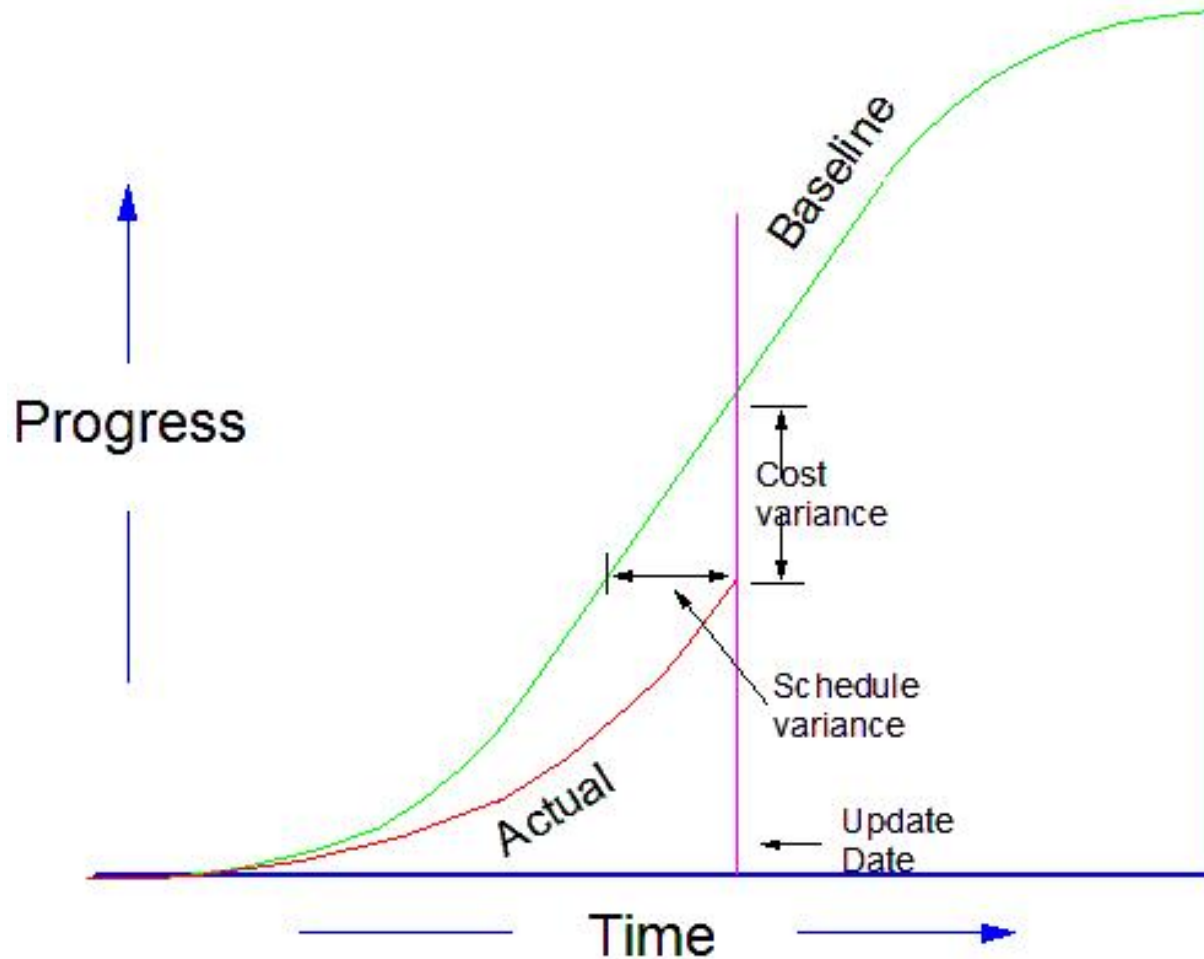
Lesson Six: Execute & Close of software project

Earned Value – Cost Variance



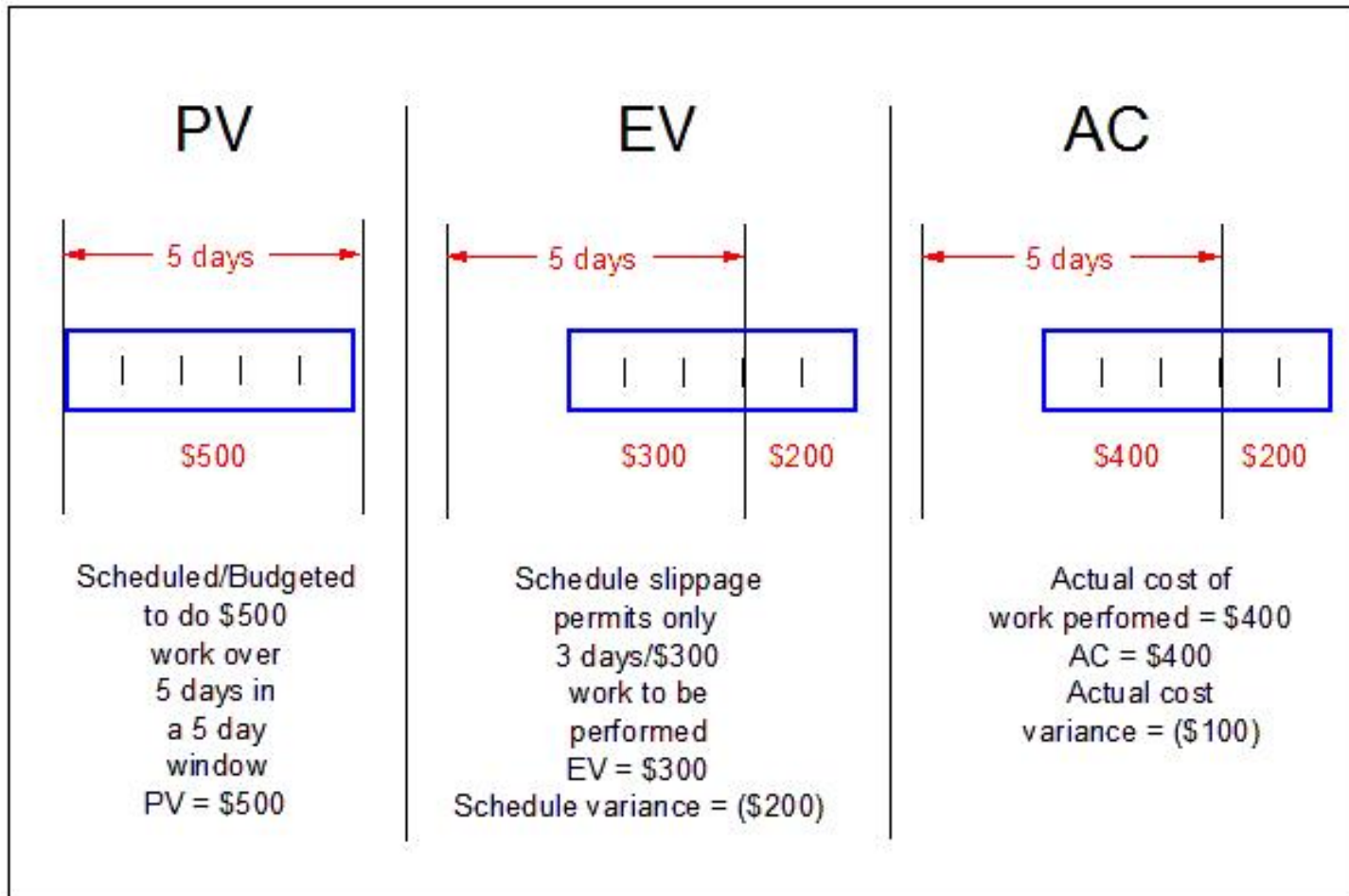
Lesson Six: Execute & Close of software project

Earned Value – Schedule Variance



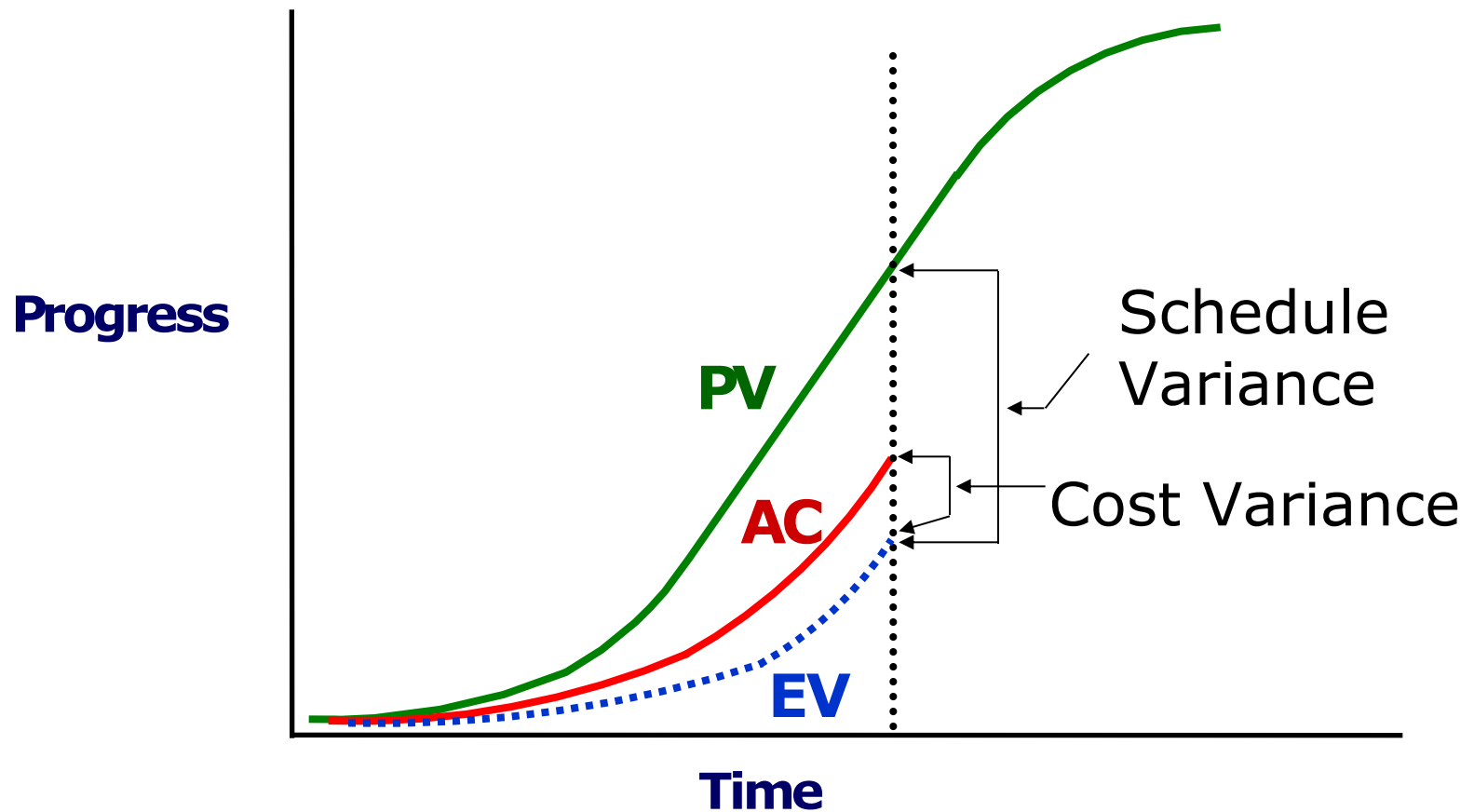
Lesson Six: Execute & Close of software project

How to Measure Earned Value



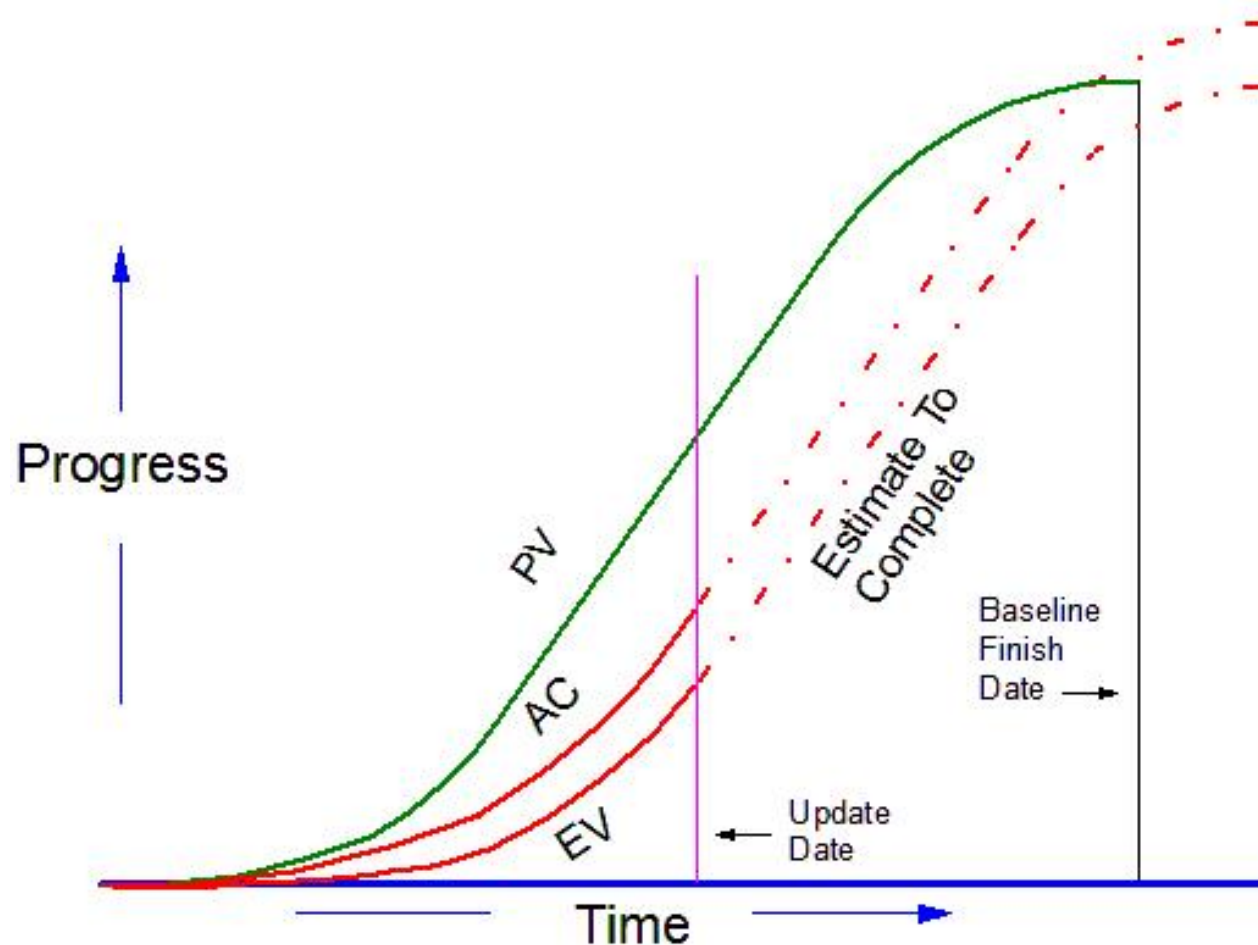
Lesson Six: Execute & Close of software project

Earned Value – The Full Story



Lesson Six: Execute & Close of software project

Earned Value – PV, EV and AC curves



Lesson Six: Execute & Close of software project

Earned Value – Basic Performance Indices

Schedule Performance Index (SPI)

A measure of how close the project is to performing work as it was actually scheduled.

$$SPI = EV / PV$$



Cost Performance Index (CPI)

A measure of how close the project is to spending on the work performed to what was planned to have been spent.

$$CPI = EV / AC$$

INDEX VALUES

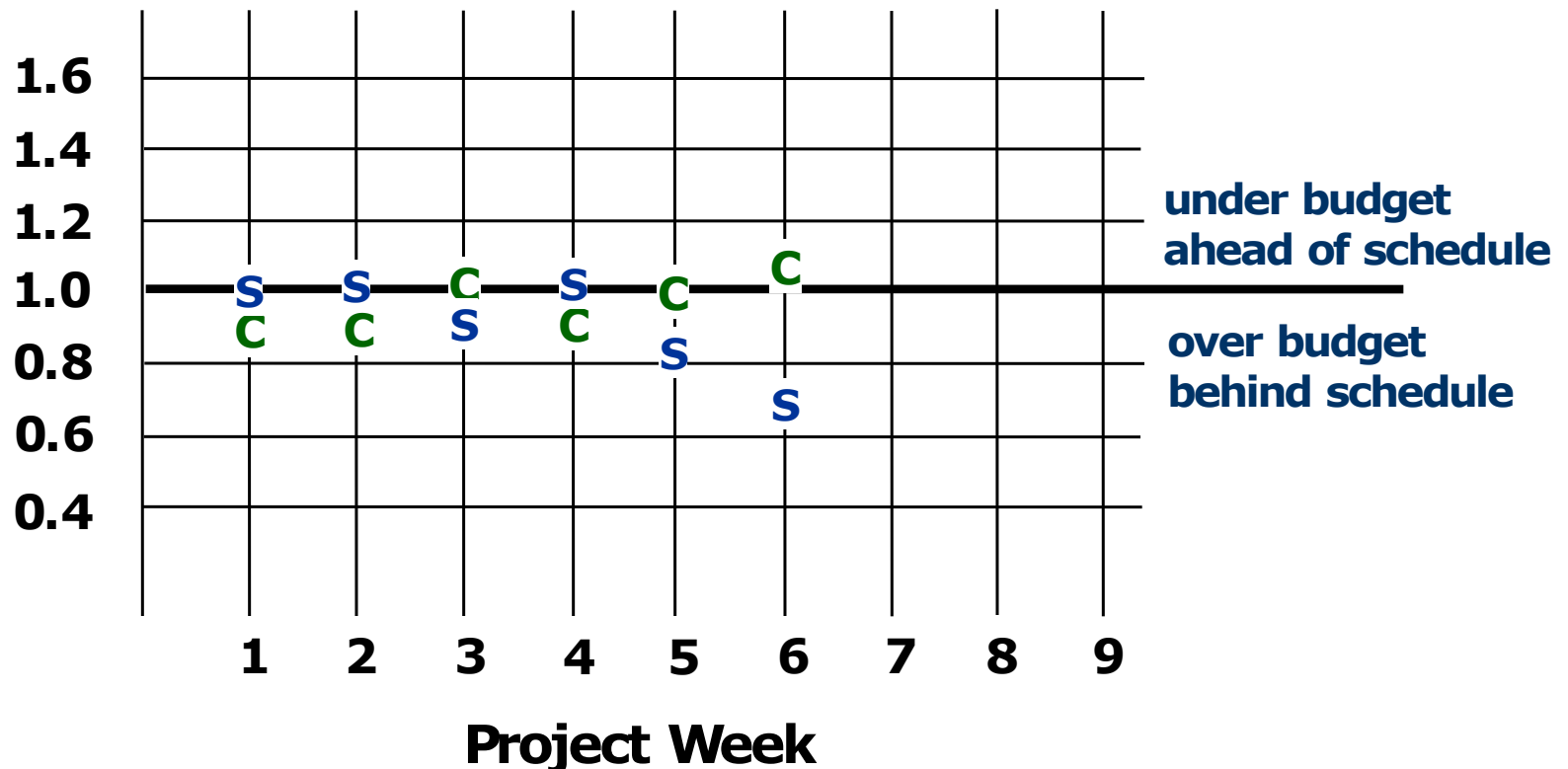
< 1: over budget or behind schedule

> 1: under budget or ahead of schedule

Lesson Six: Execute & Close of software project

Earned Value – Performance Indices

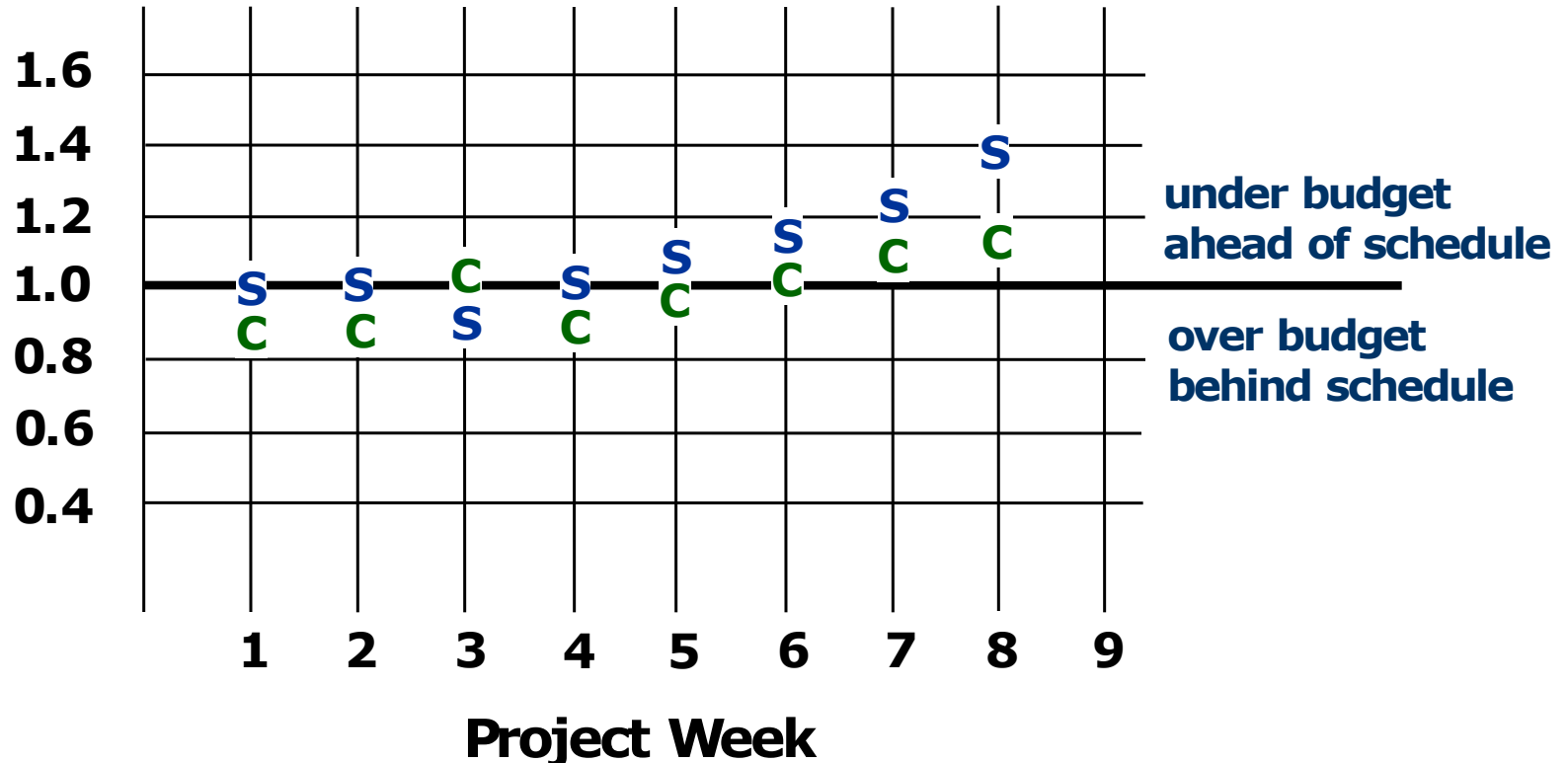
Project: ALPHA



Lesson Six: Execute & Close of software project

Earned Value – Performance Indices

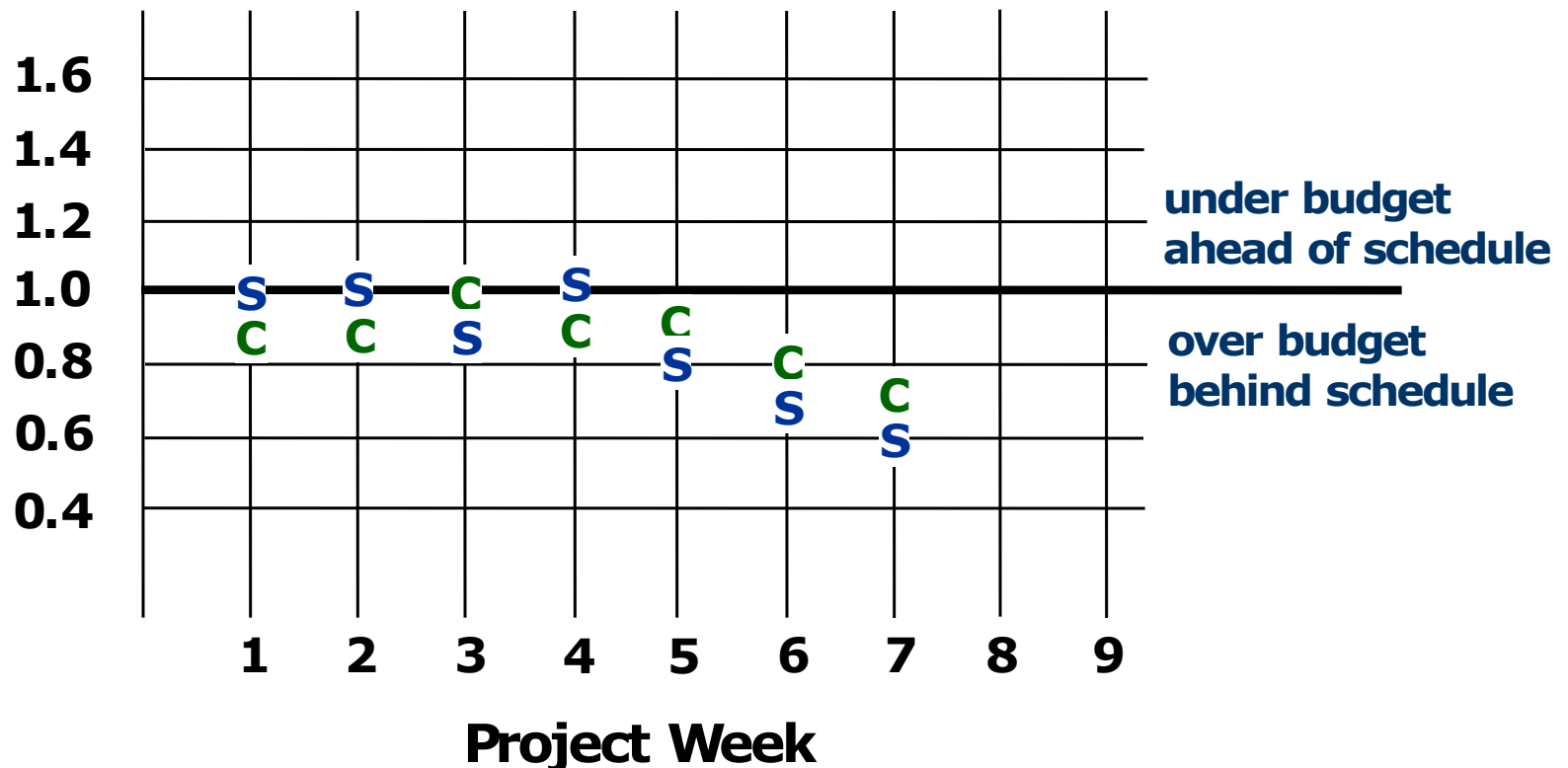
Project: ALPHA



Lesson Six: Execute & Close of software project

Earned Value – Performance Indices

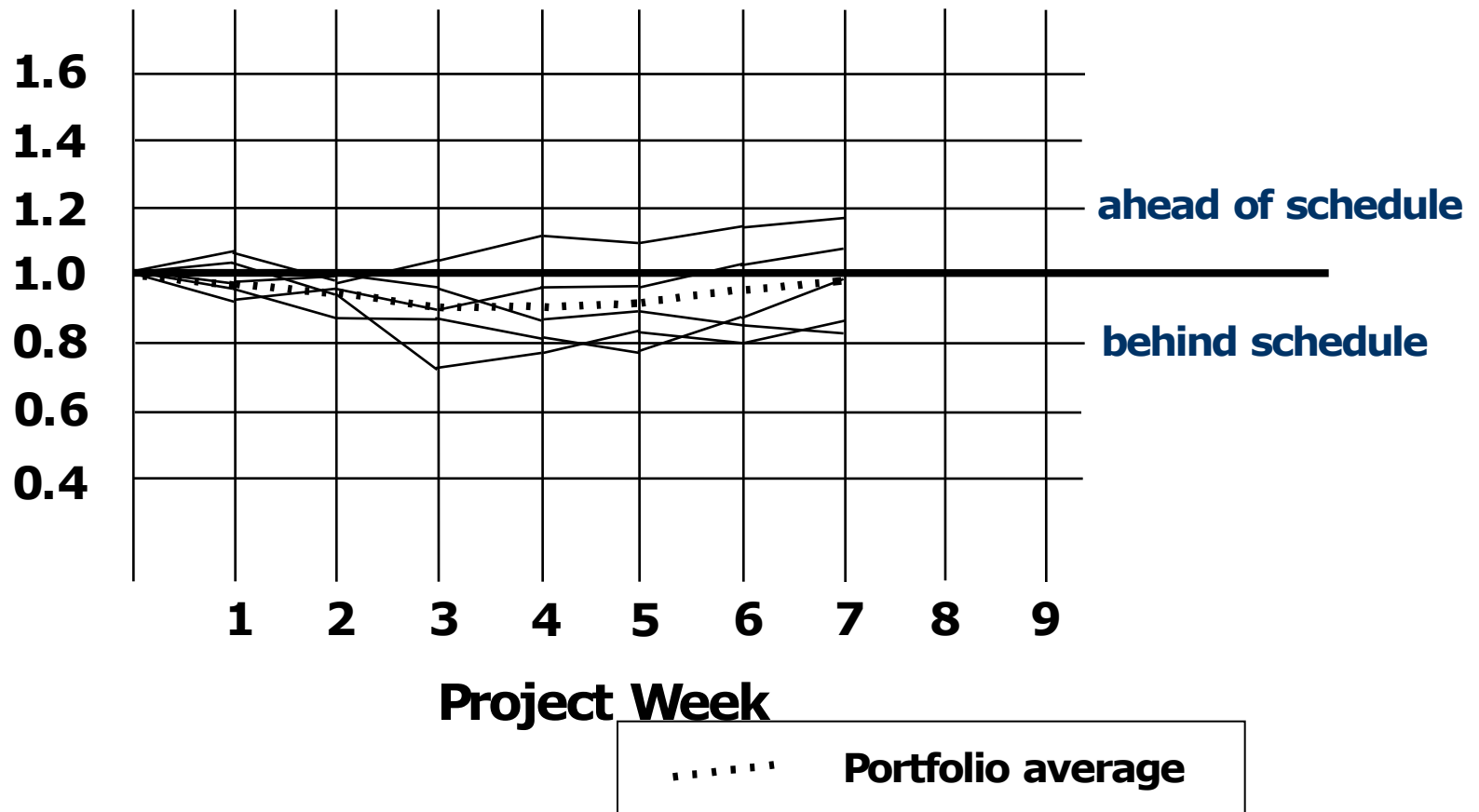
Project: ALPHA



Lesson Six: Execute & Close of software project

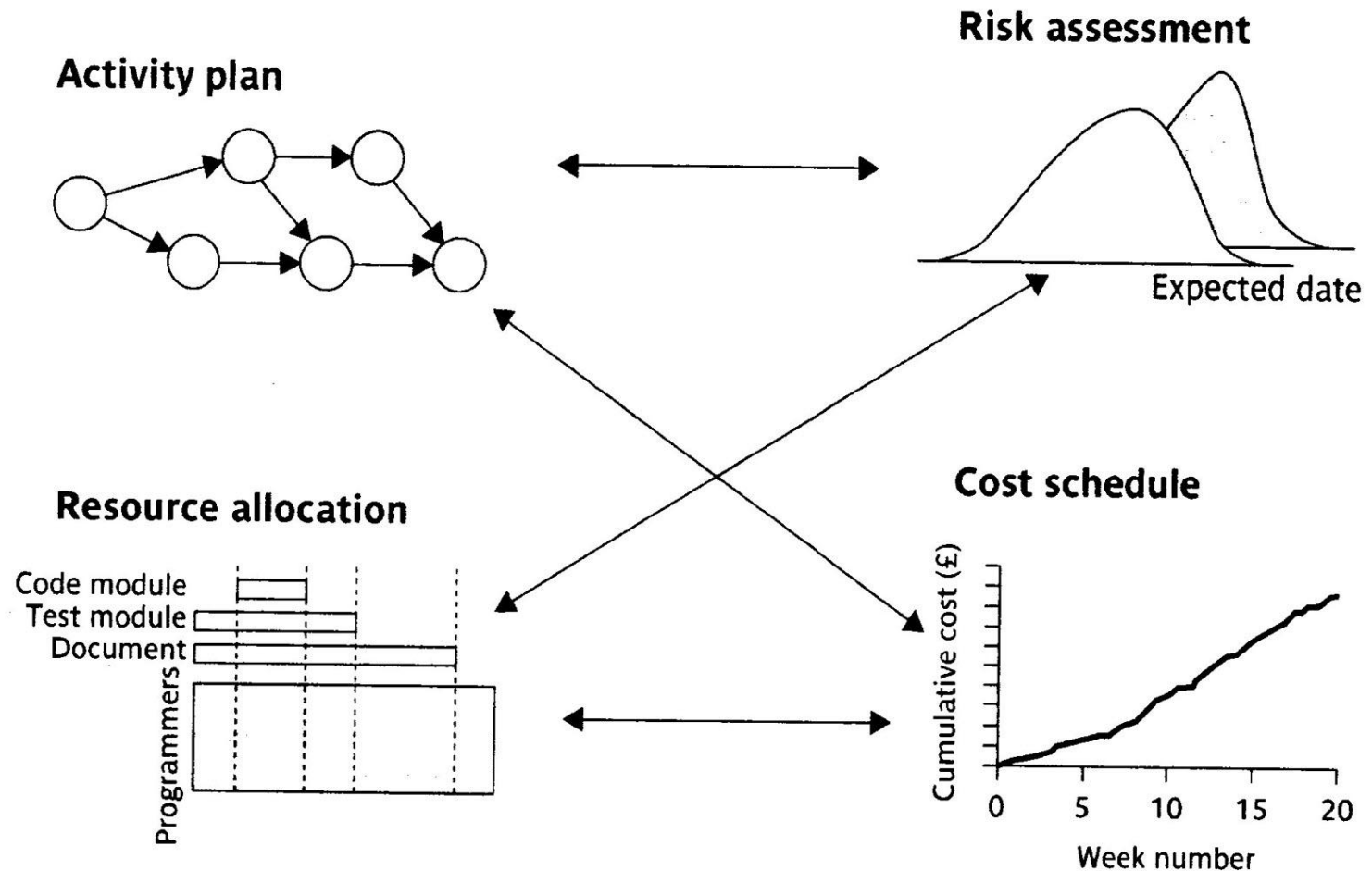
Earned Value – Performance Indices

Portfolio: BETA Program



Lesson Six: Execute & Close of software project

Example – Multiple Tools & Charts



Lesson Six: Execute & Close of software project

Managing the Scope Bank

- Initial deposit of 10% of total labor days
- All of the unfinished functions and features and the labor time to develop them are also deposited in the Scope Bank.
- The time to process and integrate a Scope Change request draws time from the Scope Bank.
- To add time to the Scope Bank remove unfinished functions and features and deposit their labor time in the Scope Bank.
- Client should continuously reprioritize contents of the Scope Bank

Lesson **Six**: Execute & Close of software project

Maintaining the Issues Log

- ID Number
- Date logged
- Description of the problem
- Impact if not resolved
- The problem owner
- Action to be taken
- Status
- Outcome

Lesson **Six**: Execute & Close of software project

Managing Project Status Meetings

- Who Should Attend?
- When Are They Held?
- What Is Their Purpose?
- What Is Their Format?

Lesson Six: Execute & Close of software project

The 15 Minute Daily Status Meeting

- Entire team or Task Managers for tasks open for work
- Everyone stands up
- Rotate the meeting facilitator
- Status of each task is reported
 - On schedule
 - Ahead of schedule (by how much)
 - Behind schedule (by how much and get well plan)
- Update Scope Bank
- Update Issues Log

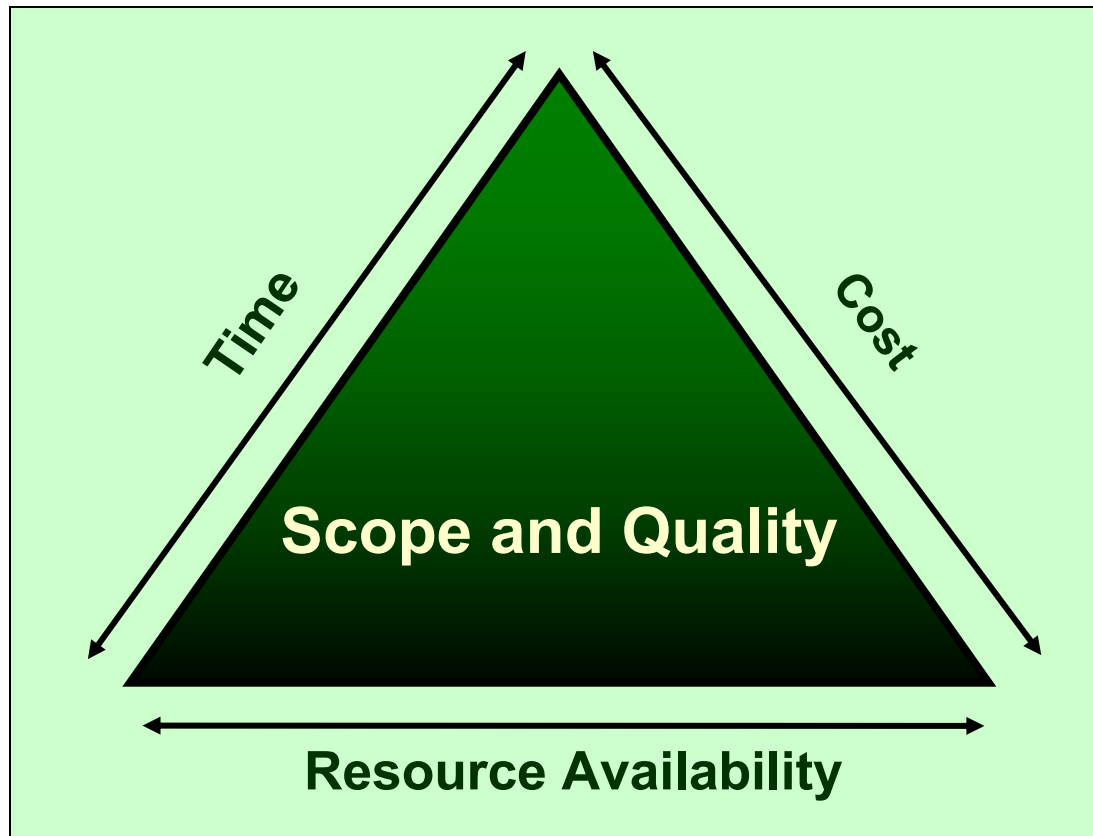
Lesson Six: Execute & Close of software project

Problem Management Meeting

- Affected parties only
- Agree on problem
- Agree on who owns the problem
- Brainstorm solutions
- Prioritize solutions
- Update Issues Log
- Schedule next meeting

Lesson Six: Execute & Close of software project

Problem Escalation Strategies – Who Controls What?



Lesson Six: Execute & Close of software project

Problem Escalation(逐级上升) Strategies

- Project Manager-Based Strategies
 - No action required. Problem will self-correct
 - Examine dependency relationships
 - Reassign resources
- Resource Manager-Based Strategies
 - Negotiate additional resources
- Client-Based Strategies
 - Negotiate multiple release strategies
 - Request schedule extension

Lesson Six: Execute & Close of software project

Escalation Strategy Hierarchy

- No action required (schedule slack will correct the problem).
- Examines FS dependencies for schedule compression opportunities.
- Reassign resources from non-critical path tasks to cover the slippage.
- Negotiate additional resources.
- Negotiate multiple release strategies.
- Request schedule extension from the client.



Lesson **Six**: Execute & Close of software project

Gaining Approval to Close the Project

When the client is satisfied that the acceptance criteria have been met the project enters the closing phase

Lesson **Six**: Execute & Close of software project

Closing a Project

1. Getting client acceptance of deliverables.
2. Ensuring that all deliverables are installed.
3. Ensuring that all documentation is in place.
4. Getting client sign-off on the final report.
5. Conducting the post-implementation audit.
6. Celebrating success.



Lesson **Six**: Execute & Close of software project

Why Do I Need Client Acceptance Procedures?

- Acceptance criteria must be clearly defined
- Criteria defined during project planning
- Avoid 11th hour disputes



1. The criteria is defined in collaboration with the client.



2. The criteria is administered by the project team during final acceptance testing.

Lesson **Six**: Execute & Close of software project

Documenting the Project

- Reference for future changes in deliverables.
- Historical record for estimating duration and cost on future projects, activities and tasks.
- Training resource for new project managers.
- Input for further training and development of the project team.
- Input for performance evaluation by the functional managers of the project team members.

Lesson **Six**: Execute & Close of software project

Documenting the Project – The Project Notebook

- POS
- Proposal and backup data
- Original and revised project schedules
- Minutes of all project team meetings
- Copies of all status reports
- Design documents
- Copies of all change notices
- Copies of all written communications
- Outstanding issues reports
- Final report
- Sample deliverables (if appropriate)
- Client acceptance documents
- Post-implementation audit report

NOTE:
Start the project notebook on the first day

Lesson **Six**: Execute & Close of software project

Conducting the Post-Implementation Audit

- Was the project goal achieved?
 - Does it do what project team said it would?
 - Does it do what client said it would?
- Was the project work done on time, within budget, and according to specification?
- Was the client satisfied with the project results?
- Was business value realized?
 - Check success criteria
- What lessons were learned about your project management methodology?
- How well did the team follow the methodology?

Lesson **Six**: Execute & Close of software project

Reasons for not doing a post-implementation audit

- Managers don't want to know.
- Managers don't want to pay the cost.
- It's not a high priority.
- There's too much other scheduled work to do.

Lesson **Six**: Execute & Close of software project

Final Project Report

- Executive Summary
- Overall success and performance of project
- Organization and administration of project
- Techniques used to accomplish results
- Strengths and weaknesses of the approach
- Recommendations
- Appendices
 - POS
 - WBS
 - Resource Schedule
 - Change Requests
 - Final Deliverables
 - Other

Lesson Six: Execute & Close of software project

Project Management Process

Initiating

Process: **Develop project charter**

Output: Project charter

Planning

Process: **Develop project management plan**

Output: Project management plan

Executing

Process: **Direct and manage project execution**

Outputs: Deliverables, work performance information, change requests,
project management plan updates, project document updates

Monitoring and Controlling

Process: **Monitor and control project work**

Outputs: Change requests, project management plan updates,
project document updates

Process: **Perform integrated change control**

Outputs: Change request status updates, project management plan updates,
project document updates

Closing

Process: **Close project or phase**

Outputs: Final product, service, or result transition;
organizational process assets updates

Project Start

Project Finish

Lesson Six: Execute & Close of software project

Tools, Templates, & Processes Used to execute project

- Current period reports
- Cumulative reports
- Exception reports
- Stoplight reports
- Variance reports
- Gantt charts
- Burn charts
- Milestone trend charts
- Earned value analysis (EVA)
- Integrated milestone trend charts and EVA
- Project status meetings
- Problem escalation strategies

Lesson Six: Execute & Close of software project

Summary of Lesson 6

- Establishing your progress reporting system.
- Applying graphical reporting tools.
- Building and maintaining the Issues Log.
- Managing project status meetings.
- Writing & maintaining client acceptance procedures
- Closing a project.
- Getting client acceptance.
- Documenting the project.
- Conducting the post-implementation audit.
- Writing the final report.

Lesson Six: Execute & Close of software project

Assignment of Lesson Six

- Each team should design and write the **Project Charter** based on the characteristics of your course project.
- The contents of **Project Charter** should be written according to the template on the textbook.
- The team leader should submit **Project Charter** of your team on canvas platform before **29th March, 2026**.